

## USE CASE 4 : AI-Assisted Code Review & Quality Governance

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### Covers:

- AI + Code Review Mindset
  - SonarQube
  - Readability
  - Complexity
  - Hallucinated APIs
  - Licensing risk
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### Scenario:

Before deployment, AI reviews the code.

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### The AI Must:

- Detect:
    - Missing null checks
    - High complexity
    - Hardcoded secrets
    - Fake APIs
  - Check license risk of dependencies
  - Suggest refactoring
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### Example Prompt:

"Review the following Order Service code.

Checklist:

- Code coverage issues
- Cyclomatic complexity
- Security flaws
- Hallucinated APIs
- Licensing concerns

Output in structured JSON:

```
{
  security_issues: [],
  complexity_score: "",
  refactor_suggestions: [],
  risk_level: ""
}
```

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**What to Build:**

- Run SonarQube scan
- Compare AI review vs Sonar
- Fix detected issues
- Document differences

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**Deliverables:**

- AI review report
- SonarQube report
- Refactored code
- Quality improvement summary